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Learner-centered design, interdisciplinarity, collaboration, and systems approaches to prepare students to address complex (i.e., multifaceted, dynamic, open), global challenges.

My teaching philosophy is shaped by industry, K-12, and higher education experience and grounded in research on how people learn and become STEM practitioners. My vision reflects national calls for student-centered, competency-driven STEM education¹, integrated technical and professional learning^{2,3}, and convergence-oriented education (i.e., approach involving deep integration of disciplines and a focus on specific, compelling problems) fostering collaborative problem solving⁴⁻⁶.

My philosophy is also informed by my personal professional trajectory. Raised in New Orleans, I witnessed engineers restore power, ensure clean water, and repair damaged infrastructure. I learned STEM is built on relationships, place-based, and collaborative, and effective solutions emerge from effective systems and work across disciplinary, institutional, and social boundaries. My professional experience has reinforced this. I trained as an engineer, worked as an engineer on energy projects worldwide, and coordinated engineering teams across technical and cultural contexts. In these settings, I observed significant variation in how colleagues navigated collaboration, problem framing, and learning in practice. I started to ask why differences existed, and how education systems might answer these questions during my doctorate.

Teaching Principles: My teaching recognizes *students as whole persons whose knowledge, identities, and experiences are essential resources for their learning and motivation for it*⁷⁻¹⁰. I prepare students for STEM workforce realities with active, reflective, and situated learning^{3,11-13}. My principles include active learning through reflection^{14,15}, co-constructive or cooperative models for learning¹⁶⁻¹⁸, contextualized engagement^{11,19-21}, interdisciplinary competency development²²⁻²⁶, and systems thinking and design approaches²⁷⁻³².

Active Learning: Reflective Practice, Co-Construction, and Situated Engagement

I design learning for students to actively *construct knowledge together (i.e., co-construction), engage with problems situated in real communities and contexts, and to reflect on their learning*. These practices are aligned with a Community-of-Practice (CoP) model^{33,34}, known to deepen student understanding and application, as well as to support transfer across domains. In CoPs, groups convene around shared learning goals, working and developing together to deepen their overall practice^{35,36}. Despite their promise, CoPs also require structured support and can be difficult to sustain. As an example CoP-informed practice, as a TA in STEM Intercultural Communication, I facilitated fishbowl discussions where an inner circle of students discussed assigned readings first while an outer circle observed, followed by a structured role switch to support inclusive participation and a shared approach to building understanding. I also served as a consultant and judge for place-based research projects where these students worked in groups to redesign accessible common application interfaces based on community feedback. I led sessions on communicating technical ideas to diverse audiences prior to structured judging, but the active learning with students as designer-communicators helped them take on understanding beyond technical outputs in ways lecturing alone could not³⁷. Critically, students did not always immediately buy into these active practices^{38,39}; we had to scaffold them, set explicit expectations for, and justify their presence in our course to ensure students benefitted.

My commitment to co-construction and situated community engagement is also reflected in my participatory action research supporting digital literacy for newcomers and hospital educators in Virginia⁴⁰⁻⁴³. As a PhD student, I managed multiple interdisciplinary undergraduate research teams partnered with community organizations to co-develop (with community members) learning applications, websites, and databases of resettlement services. This experience exposed students to real-world application and communities directly connected to their STEM expertise, but in a format not typically embedded in curricula²¹. We have published on this approach as a pedagogical practice⁴⁰ and are developing an action-oriented course for engineers building on outcomes. Studio-based pedagogies, increasingly adopted in STEM, represent promising avenues for sustaining this at classroom scale⁴⁴.

The situated learning I promote also requires active reflection²⁰ and writing to learn^{45,46}, rounding out this first teaching principle. I incorporate structured reflective writing practices that move beyond rote recall of engineering facts toward student metacognition, connection-making, and reasoning about how engineering implicates surroundings or how one courses' content connects to others. Drawing on colleagues' research^{19,47} and scholarship on writing as active learning pedagogy for critical thinking⁴⁸, I design assignments to prompt students to examine how their thinking evolves as they engage with new ideas, connect across boundaries, and relate knowledge to prior experience. This practice deepens understanding of

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technical concepts as they are related to their own lives^{12,13}. Rather than relying exclusively on traditional discussion posts, I prefer to use situated creative assignments like group creation of Wikipedia pages⁴⁹ about STEM case studies because they require collaborative student synthesis, audience consideration, *and* design reflection, all critical professional skills that are not a hallmark of traditional lecture-based teaching alone.

I am also attentive to the sustainability of these active learning practices. While innovative pedagogies are essential for student learning, they are demanding to maintain^{50,51}. I prioritize approaches like simple discussion structures and manageable assignment switch-outs that are pedagogically effective *and* scalable, supporting long-term integration.

Fostering Interdisciplinary Competencies: Interdisciplinarity and collaboration are additional features of my teaching, reflecting the realities of contemporary STEM practice and research on how students learn to work together across boundaries⁵²⁻⁵⁴. I design learning help students make connections, integrate ways of knowing, and collaborate across disciplinary contexts. As a TA of Interdisciplinary Disaster Resilience, I designed activities where grad students worked in interdisciplinary teams to formulate convergent research studies and develop resilience strategies for real cases, aligned with case methods for teaching^{55,56}. This required navigation of differing expertise, methods, and constraints towards intended learning outcomes addressing both technical and social elements of disasters. By grounding projects in real scenarios rather than abstract ones, students had to engage with the interconnected and constrained nature of sociotechnical challenges^{57,58}. They practiced making decisions under uncertainty, negotiating trade-offs, and communicating across boundaries, skills mirroring the realities of modern STEM work^{3,5,6} compared to textbook problem sets³⁷.

Systems Approaches to Learning and Design: Systems thinking – which can be defined as “a critical interdisciplinary skill that describes the cognitive flexibility needed to collaboratively work on problems facing society”³² – is the last piece of my philosophy. STEM practitioners rarely solve isolated problems; they work within interconnected and multilayered systems. A pumping system can be analyzed at the micro level (component behavior like pump efficiency and fatigue), the meso (interactions among pumps, pipes, controls), the macro (citywide performance, demand), and the exo level (regulatory, capacity, and community trust). In my teaching, I require students to reason across these levels, moving beyond individual components to understand how technical decisions propagate across systems and shape societal perceptions.

I also extend this systems lens to how I design and evaluate education experiences⁵⁹⁻⁶⁴. I view assignments, courses, curricula, and institutions as interconnected pieces of systems embedded within larger systems like policy agencies, informing my approach to curriculum alignment, assessment, and program improvement. For instance, in supporting Rice ABET reaccreditation and lab revisions, I worked with faculty to align learning outcomes with authentic practice, remove misaligned tasks, and ensure ongoing effort advanced core goals in broader curriculum via regular updates in the future. I did this by auditing labs against stated learning outcomes (in the Rice curriculum and from ABET²) and redesigning experiments to foreground prediction and application beyond simple reflected understanding. In a strain gauge lab, this meant providing prewritten code and re-designing so students focused on interpreting gauge responses and applying understanding to prototype devices (i.e., bending beam sensors). We also implemented student surveys to serve ongoing improvement feedback loops, aligned with organizational systems practices.

Systems thinking impacts both my teaching and educational administration, providing a framework for preparing students for technically sound, contextually informed, *and* socially responsible STEM practice, while also ensuring interventions are right-sized for institutions. Instead of assuming one-size-fits-all teaching will be fruitful or implementing change misaligned with existing practice, I use systems to analyze learning, motivation, policy, and practice (and influences on them) to make data-driven decisions supporting sustained evidence-based practice. For instance, in my role as a research assistant with VT's Graduate and Professional Studies Office, I used a systems lens to create a tool helping audit course enrollment after identifying a key structural gap: grad coordinators handled registration tracking, but program directors (GPDs) were accountable for under-enrollment. This misalignment meant departments often failed to meet minimum enrollment requirements, with audits left to the wayside. Our tool automated the audit and positioned GPDs to directly address issues, enhancing motivation to comply. The tool was designed to mesh with and connect existing processes and bottlenecks, applying systems engineering design to educational contexts most beneficial to the entire system – not just students or faculty or individual units alone.

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